

Name:

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P525/1

CHEMISTRY

Paper 1

2024

$2\frac{3}{4}$ Hours

Uganda Advanced Certificate of Education

EOT II EXAMINATIONS

S.5 CHEMISTRY

Paper 1

2Hours 45 minutes

INSTRUCTIONS:

Answer **all** questions in section **A** and **six** questions from section **B**.

All answers must be written in the spaces provided.

The Periodic Table, with relative atomic masses is attached at the end of the paper.

Mathematical tables (3 - figure tables) are adequate or non - programmable scientific electronic calculators may be used.

Illustrate your answers, with equations where applicable.

Where necessary, use the following.

- Molar gas constant, $R = 8.31 \text{ kJ mol}^{-1}$
- Molar volume of a gas at s.t.p is 22.4 liters.
- Standard temperature = 273K
- Standard pressure = 101325 Nm^{-2}

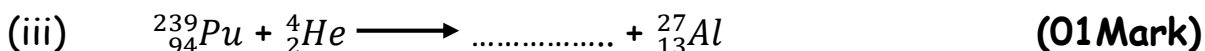
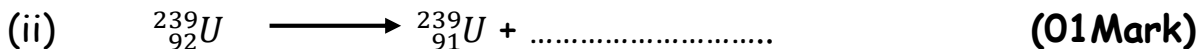
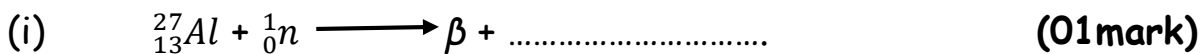
For Examiner's Use Only

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Total

SECTION A (46 MARKS)

Answer **all** questions in this section.

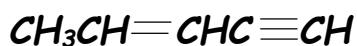
1. a) Complete the following nuclear equations.



b) The half-life of a radioactive element X is 36 years. After how many years will the element take to decay to 25%. (02Marks)

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2. An organic compound R, has the structure.



a) Name the functional groups present in compound R. (01 mark)

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b) Write equation for the reaction between R and

(i) Alkaline potassium manganate (VII) solution. (01Mark)

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(ii) Ammoniacal silver nitrate solution. (01Mark)

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.....
c) State what would be observed in (b) (i) and (ii) (02Marks)

(i)

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(ii)

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3. (a) Write an equation for the reaction between hot concentrated sodium hydroxide solution and

(i) Aluminium Oxide. (1½ Marks)

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(ii) Silicon. (1½ Marks)

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(iii) Phosphorous (1½ Marks)

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(iv) Chlorine (1½ Marks)

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4. (a) What is meant by **relative abundance**? (01Mark)

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b) Bromine has two naturally occurring isotopes with isotopic masses and relative abundances as shown below.

Isotopic mass	Relative abundance %
79	50.5
81	49.5

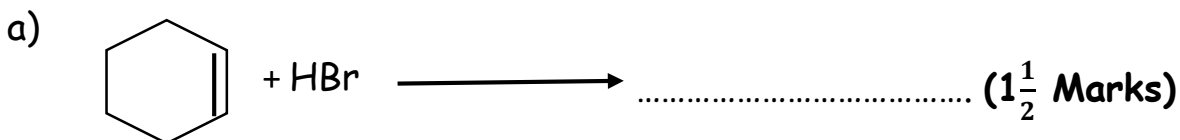
- (i) Bromine has two isotopes, **Br -79** and **Br - 81**. Write the possible ions formed on the mass spectrum when the two isotopes combine. (01Mark)

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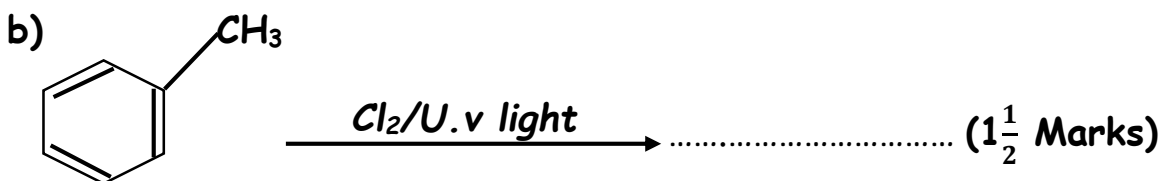
- (ii) Calculate the average atomic mass of bromine. (03marks)

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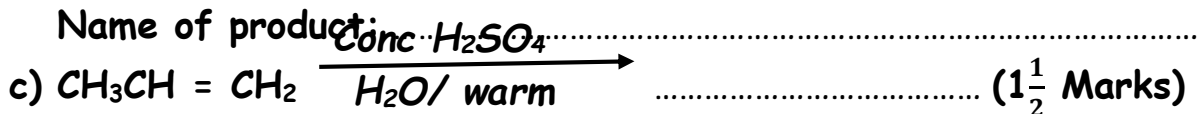
5. Complete the following organic reactions and name the major organic product formed in each case.



Name of product:



Name of product:



Name of product:

6. (a) Define diagonal relationship. (01 Mark)

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(i) State **two** properties in which beryllium resembles Aluminium. (02Marks)

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(ii) State **two** reasons why beryllium resembles Aluminium. (1mk)

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b) State any other pair of elements in the Periodic Table that exhibit diagonal relationship. (01 Mark)

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7. Draw the structure and name the shape of each species given in the table below: (05Marks)

Species	Structure	Shape
BF ₃		
H ₂ O		

ClO_2^-		
CH_3Cl		
SO_4^{2-}		

8. (a) State two characteristics of a primary standard. (02Marks)

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b) State with a reason why the following substances are not considered as a primary standard.

(i) Concentrated Sulphuric acid. (01 Mark)

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(ii) Sodium thiosulphate - 5 - water. (02Marks)

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9. (a) State what would be observed and write equation for the reaction that would take place if:

- (i) Excess hydrogen peroxide was added to an acidified solution of potassium manganate (VII). (02Marks)

Observation:

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Equation:

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- (ii) But - 1 - ene was added to a solution of bromine in tetrachloromethane. (02Marks)

Observation:

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Equation:

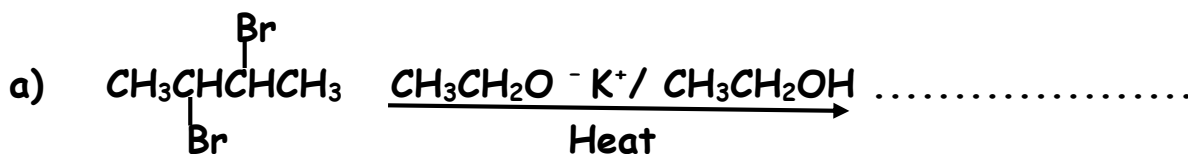
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- b) Give a reason for your observation in (a) (i) above. (01Mark)

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10. Complete the following equations and write a mechanism for the reaction that occurs.

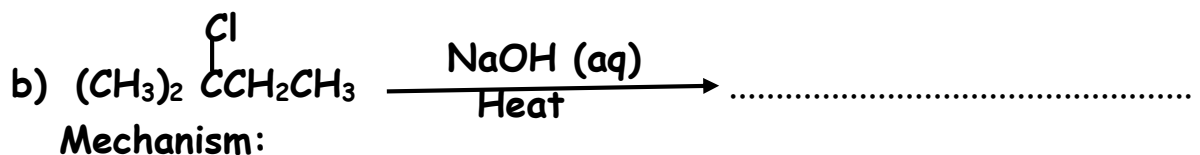


Mechanism:

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 (03Marks)



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SECTION B (54MARKS)

Answer any **six** questions from this section.

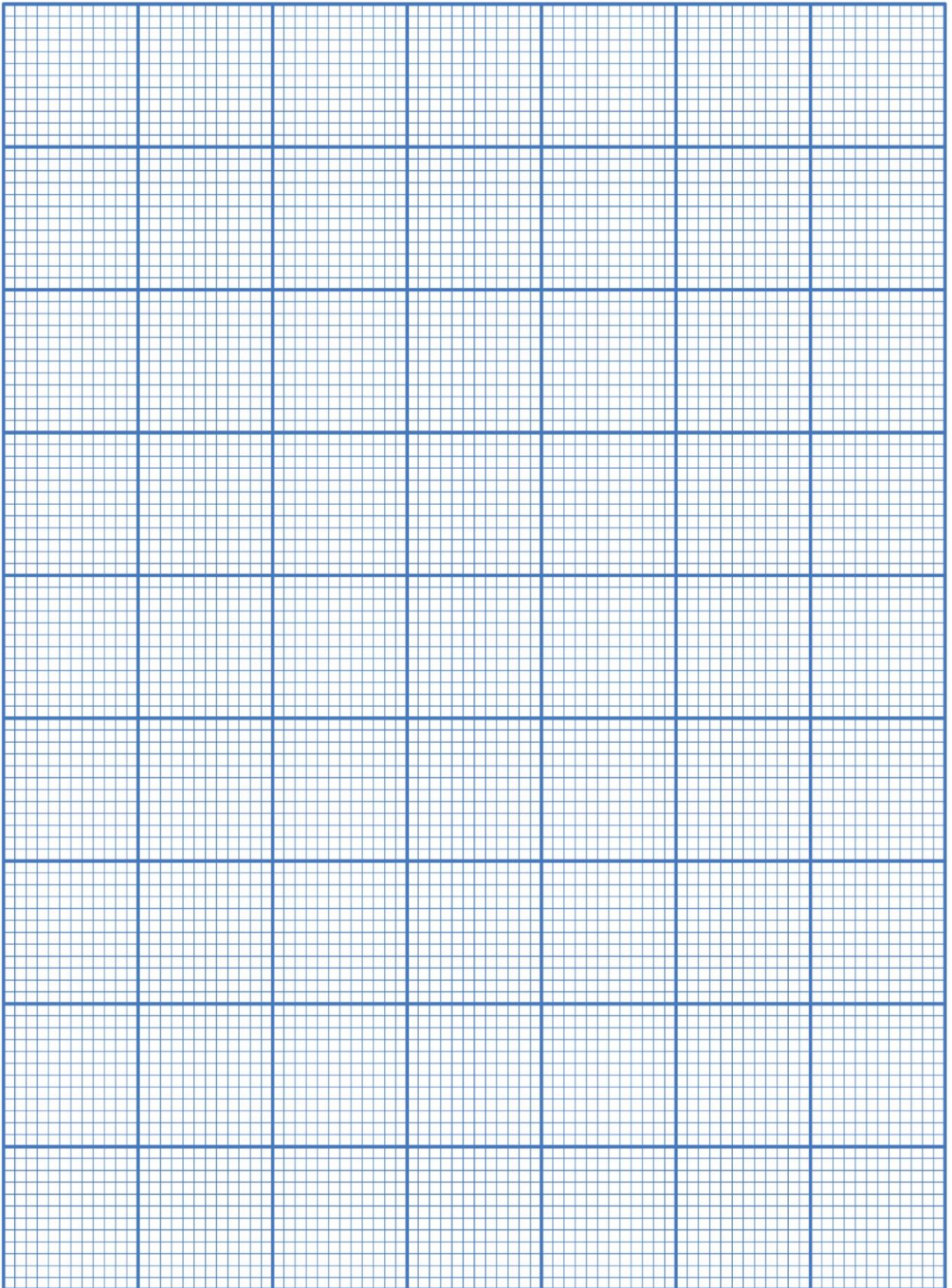
11. (a) State what is meant by the term **freezing point constant** of a substance. (01Mark)

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- b) The table below shows the freezing points of various concentrations of a non - volatile solute Q in water at 760mmHg.

Concentration of Q (gdm ³)	0	30	60	90	120	150
Freezing point/ °C	0	-0.16	-0.32	-0.49	-0.65	-0.81

Plot a graph of freezing point depression against concentration of Q, on a graph paper. (03Marks)



c) Determine the:

(i) Slope of the graph you have drawn in (b) $(1\frac{1}{2} \text{ Marks})$

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(ii) Relative molecular mass of Q (K_f of water is $1.86^\circ\text{Ckg}^{-1}\text{mol}^{-1}$) $(2\frac{1}{2} \text{ Marks})$

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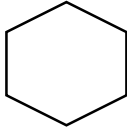
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12. Write equations to show how the following compounds can be synthesized. Indicate the condition(s) for the reaction(s)

a) $\text{HC} \equiv \text{CH} \longrightarrow$  $(2\frac{1}{2} \text{ Marks})$

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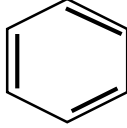
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b) $\text{H}_2\text{C}=\text{CH}_2 \longrightarrow$  $(3\frac{1}{2} \text{ Marks})$

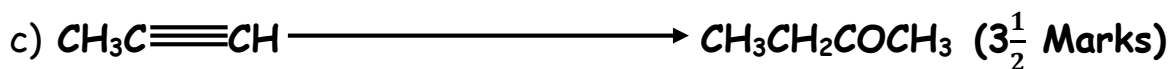
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13. 1.363g of compound Y containing carbon, hydrogen and bromine on complete combustion gave 1.10g of carbon dioxide and 0.45g of water. 0.35g of Y was vaporized, and it occupied 39.5cm^3 at 20°C and 750mmHg. Calculate the:

(a) (i) Empirical formula of Y. ($2\frac{1}{2}$ Marks)

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(ii) Molecular formula of Y. (02Marks)

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b) Y forms a compound Z when treated with a mixture of potassium hydroxide solution and ethanol under reflux. Z reacts with Ammoniacal silver nitrate solution to form a white precipitate Q.

Identify:

- (i) Y($0\frac{1}{2}$ mark)
(ii) Z ($0\frac{1}{2}$ Mark)
(iii) Q($0\frac{1}{2}$ Mark)

c) Write :

- (i) an equation for the reaction between Z and Ammoniacal silver nitrate solution. (01mark)

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- (ii) the mechanism for the reaction leading to formation of Z.

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..... (02marks)

14. (a) (i) What is meant by the term **first ionization energy**?

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..... (01Mark)

(ii) Write an equation for the first ionization energy of Aluminium.

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..... (01Mark)

- (iii) State two factors that affect the value of first ionization energy.

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..... (01Mark)

b) State and explain how first ionization energy varies:

(i) Down the group. (03Marks)

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(ii) Across a given period. (03Marks)

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15. (a) A certain volume of a gaseous hydride of silicon of the formula SiH_{2n} , diffused through a narrow porous hold in 17.08 seconds. The same volume of carbon dioxide diffused through the hole under the same conditions in 20 seconds.

(i) Calculate the relative molecular mass of the hydride.
(2½ marks)

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(ii) Determine the molecular formula of the hydride. (1Mk)

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b) (i) Explain why silicon hydride undergoes hydrolysis in the presence of sodium hydroxide solution whereas carbon hydride does not. (04Marks)

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(ii) Write an equation for hydrolysis of silicon hydride in sodium hydroxide solution. ($1\frac{1}{2}$ Marks)

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16. Name one reagent that can be used to distinguish between the following pairs of compounds. In each case state what would be observed if each member of the pair is treated with the named reagent.

(a) $CH_3C\equiv CCH_3$ and $CH_3CH_2C\equiv CH$ (03Marks)

Reagent:

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Observations:

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b) $\text{CH}_3\text{CH}_2\text{Br}$ and $\text{CH}_3\text{CH}_2\text{Cl}$

(03Marks)

Reagent:

Observations:

c) $\text{H}_2\text{C}=\text{CH}_2$ and CH_3CH_3

(03Marks)

Reagent:

Observations:

17. 2.40g of bronze (an alloy of copper and tin) was crushed and heated with excess concentrated Sulphuric acid and the mixture filtered. The resultant filtrate was made up to 250cm^3 with distilled water. 25cm^3 of this solution required 19.20cm^3 of 0.02M potassium manganate (VII) solution for complete reaction.

a) Write the equation(s) for the reaction(s) that took place. (3mrks)

b) Calculate the number of moles of manganate (VII) ions that reacted.

(03Marks)

c) Determine the percentage by mass of tin in the bronze. (Sn = 119)
(03marks)

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END

1	2											3	4	5	6	7	8
1 H 1.0															1 H 1.0	2 He 4.0	
3 Li 6.9	4 Be 9.0											5 B 10.8	6 C 12.0	7 N 14.0	8 O 16.0	9 F 19.0	10 Ne 20.2
11 Na 23.0	12 Mg 24.3											13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.4	18 Ar 40.0
19 K 39.1	20 Ca 40.1	21 Sc 45.0	22 Ti 47.9	23 V 50.9	24 Cr 52.0	25 Mn 54.9	26 Fe 55.8	27 Co 58.9	28 Ni 58.7	29 Cu 63.5	30 Zn 65.	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8
37 Rb 85.5	38 Sr 87.6	39 Y 88.9	40 Zr 91.2	41 Nb 92.9	42 Mo 95.9	43 Tc 98.9	44 Ru 101	45 Rh 103	46 Pd 103	47 Ag 108	48 Cd 112	49 In 115	50 Sn 119	51 Sb 122	52 Te 128	53 I 127	54 Xe 131
55 Cs 133	56 Ba 137	57 La 139	72 Hf 178	73 Ta 181	74 W 184	75 Re 186	76 Os 190	77 Ir 192	78 Pt 195	79 Au 197	80 Hg 201	81 Tl 204	82 Pb 207	83 Bi 209	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)															
			57 La 139	58 Ce 140	59 Fr 141	60 Nd 144	61 Pm (145)	62 Sm 150	63 Eu 152	64 Gd 157	65 Tb 159	66 Dy 162	67 Ho 165	68 Er 167	69 Tm 169	70 Yb 173	71 Lu 175
			89 Ac (227)	90 Th 232	91 Pa 231	92 U 238	93 Np 237	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf 251	99 Ea (254)	100 Fm (257)	101 Mv (256)	102 No (254)	103 Lw 260

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- H – indicates Atomic number**
 - H – indicates relative Atomic mass**
- 1.0